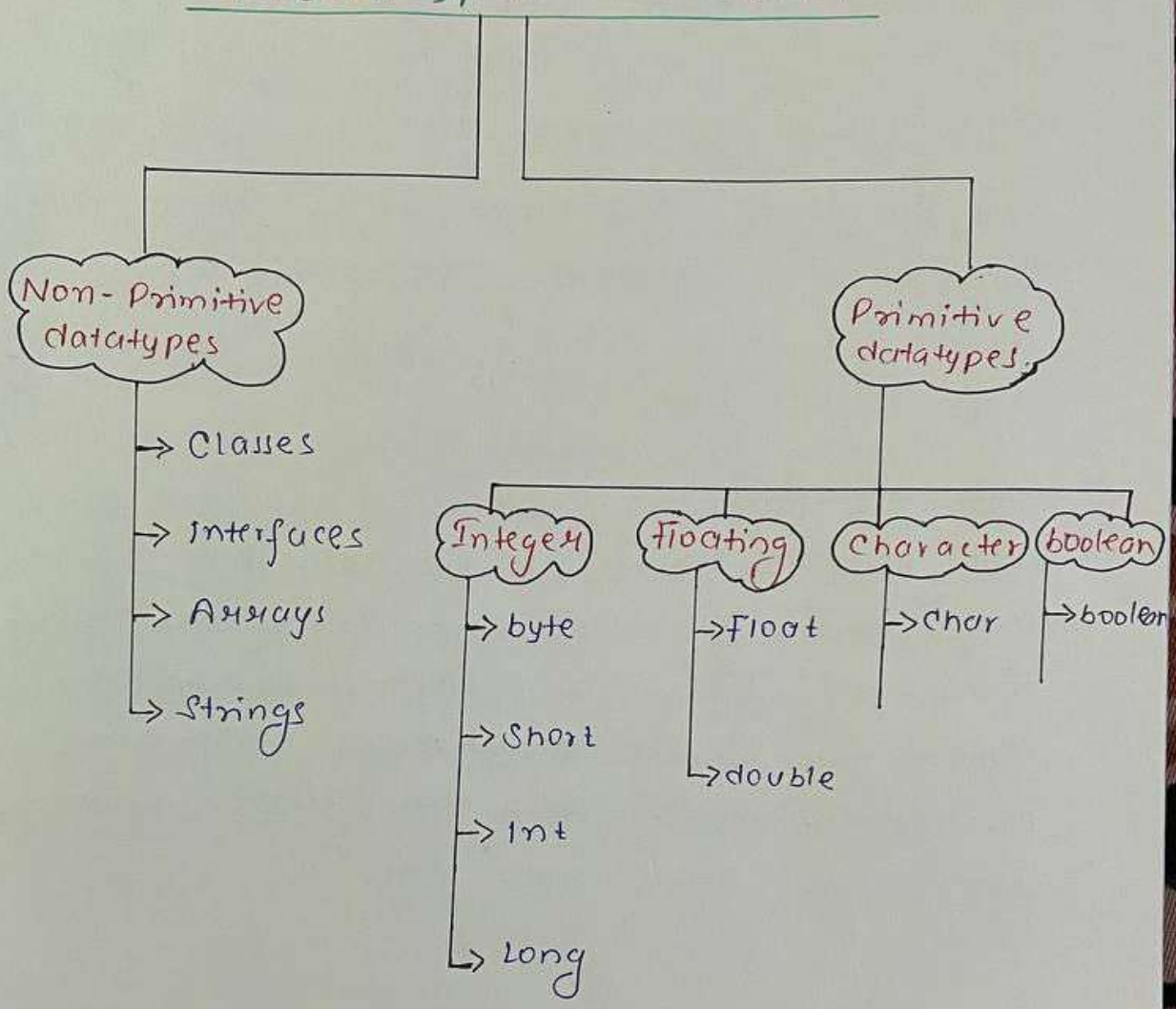
Datatypes in Java



Primitive data-types: Primitive datatypes Java ke basic built in datatypes hote hai jaise int, float, boolean etc, jo ek simple value store karte hai aur inke pass koi methods nahi hote hai.

Non-Primitive datatypes: Non-primitive datatypes Object-based hote hai jaise String, Array or custom classes. Ye reference store krte hai, inko Object heap me store hota hai, ye null bhi ho skte hai aur ye methods ke sath aate hai.

"Java me primitive data types stact memory me directly store hote hai, kyuki ye lightweight aur fixed size ka hota hai.
Non-primitive data types jaise String, Array, Class inka reference stack me store hota hai aur actual object heap memory me store hota hai."

"Java me primitive data types ka type aur size compile-time pe fix hota hai, lekin unki memory allocation stack me run-time pe hota hai. Non-primitive data types ka object heap mein run-time pe create hota hai, aur unka reference stack me store hota hai."

Type Casting in Java

Definition: Typecasting ka matlab hota hai ek datatype ke variable ko doosre data type me convert karna.

Types of Typecasting in Java.

- Widening (Implicit)
- Narrowing (Explicit)

(i) Widening Type Casting (Implicit)

Jab hum chhoti size ke type ko badi size ke type me convert karte hain. Java automatically kar deta hai, Data ka loss nahi hota hai.

Order of data type (smaller to bigger).

byte < short < int < long < float < double

ex:- int a = 10

double b = a // automatic typecasting (int to double).

(ii) Narrowing Typecasting (Explicit)

Jab hum bade type ko chhote type me convert karte hain.

Jave me iske liye manually cast karna padta hai, kyunki data lost ho sakta hai.

ex:- double a = 9.78;

int b = (int) a; // Explicit Casting

System.out.println(b); // output : 9 (0.78 lost)

Special Type Casting.

(i) char -> int (widening, Implicit Casting)

Java automatically char ko int me convert kar deta hai, kyonki char ka value internally Unicode number (asci value) hota hai.

```
example: char ch = 'A';
         int num = ch;
         System.out.println(num); // output: 65
```

(ii) int -> char (Narrowing, Explicit or Implicit if safe)

Jab int ko char me convert karte hain to JVM Unicode table ke according character return karta hai.

```
ex:- int num = 66;
     char ch = (char) num;
     System.out.println(ch); // output: B.
```

66 ka Unicode character 'B' hota hai, to conversion successful hai.

NOTE: Agar int value 0 - 65535 ke range me hai, to cast safe hai.

```
ex:- int num = 70000;
     char ch = (char) num;
     System.out.println(ch); // output: weird character
```

Yaha 70000 char ke valid range (0-65535) se bahar hai, to result unexpected hoga.